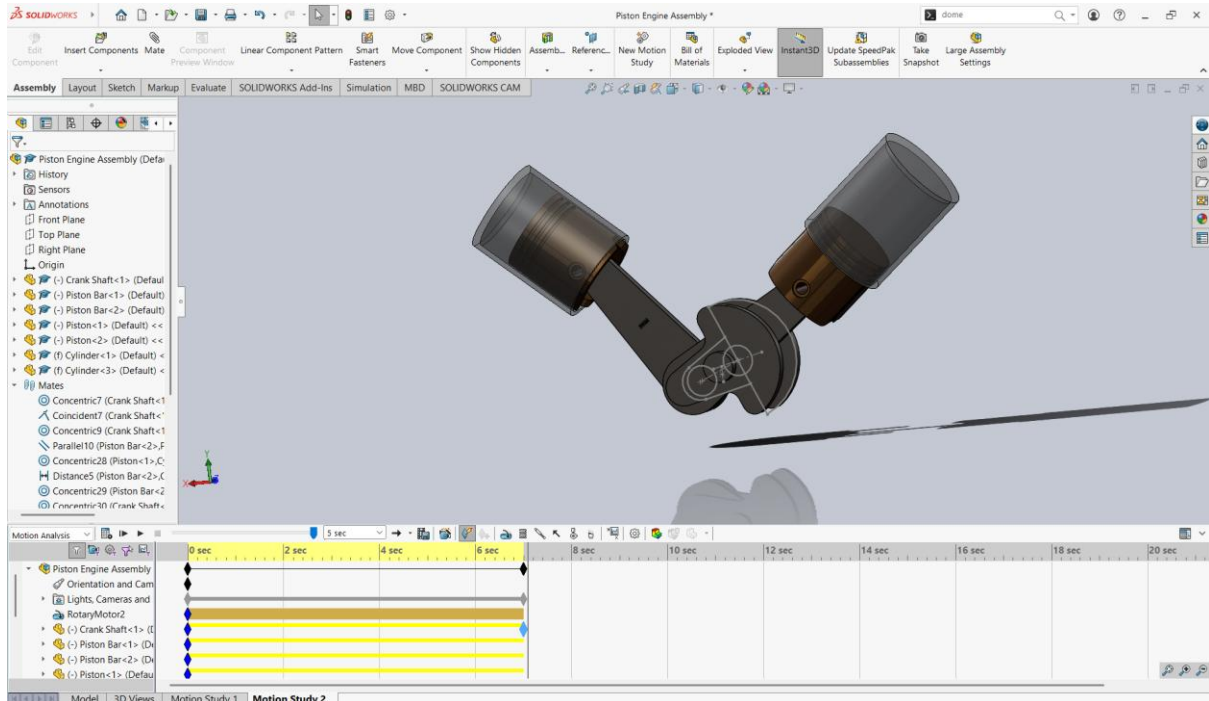


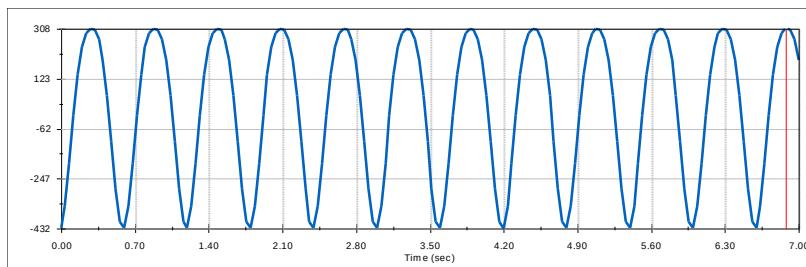
3B5 Mechanics of Machines – Assignment 1

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Linear Velocity In the Y-Direction:



The above images show the motion study of the SolidWorks assembly, and a linear velocity graph in the y-direction. The mechanism is the basis of a two-cylinder piston engine, connected to a crank shaft and firing into 2 cylinders. This type of engine is commonly seen in motorbikes or small vehicles and is often referred to as a v-twin engine. When one of the pistons is at top dead centre, the other is at bottom dead centre. This ensures that the engine is balanced when in operation.

All of the components in this assembly were self-made, with some inspiration from some videos on YouTube.